System One Models Are Not Better Rerankers

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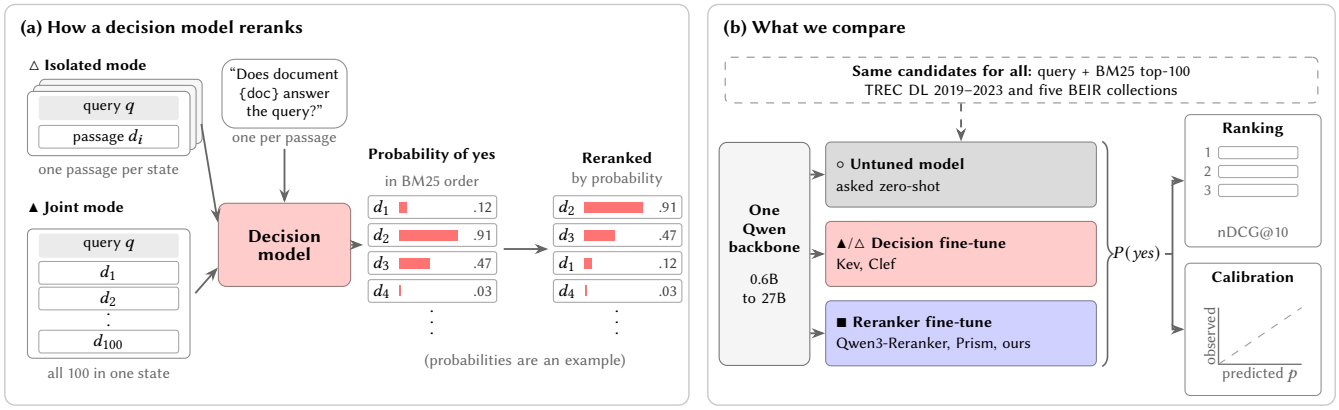


Figure 1: How System One models rerank and what we compare. (a) A decision model is asked, for each BM25 candidate, whether the passage answers the query, and the candidates are sorted by the probability of "yes". The state, the input the model is asked about, holds one passage (isolated mode) or all 100 candidates (joint mode). (b) On one Qwen backbone, the untuned model, its decision fine-tunes and its reranker fine-tunes rerank the same candidates and are compared on ranking and on calibration.

Abstract

System One models, or decision models, answer a question about an input with a calibrated probability instead of generated text. Within weeks of the first release, in September 2026, they were used as rerankers. We test whether the decision fine-tuning behind them makes a better reranker than plain reranker fine-tuning of the same model. For six Qwen backbones from 0.6B to 27B parameters and ten TREC DL and BEIR collections, we compare the untuned model, asked zero-shot, with its decision fine-tunes and its reranker fine-tunes, two of which we train ourselves. Decision fine-tuning never makes the better reranker. From 0.6B to 9B, reranker fine-tuning wins on every backbone by 5 to 29 points nDCG@10, also on collections that no fine-tune was trained on, and at 27B the two tie. Up to 9B, decision fine-tuning adds nothing over simply asking the untuned model. The calibration advantage of decision models depends on the question. Asked whether a passage answers the query, they are better calibrated than the published reranker fine-tunes from 4B on. Asked a looser question from prior work, they lose this advantage, and the untuned models gain as much from the wording. Nor does calibration make up for a worse ranking: up to 9B, a reranker fine-tune keeps the better passages at a fixed threshold. If you need a reranker, fine-tune a reranker.

Keywords

Reranking, Calibration, System One Models, Decision Models, Fine-Tuning, Large Language Models

1 Introduction

Cross-encoders are a well-established way to rerank the candidates of a first-stage retriever [19, 20], and open large language models fine-tuned for ranking are the current state of the art [17, 38, 39]. *System One models*, which we call decision models, promise something else. They answer a question about an input with a calibrated probability instead of generated text. The name refers to fast, intuitive thinking, Kahneman's "System 1" [14, 33]. TypeSafe AI released the first of them, Jev, on 15 September 2026, trained with an unpublished method called RLCD (reinforcement learning for calibrated decisions) [33]. Within weeks, Jev was reported to be a competitive reranker [13, 27] and was evaluated or integrated in retrieval systems [11, 15, 34]. Open decision models followed: Kev [22] and Cloudflare's Clef [4]. They are fine-tuned from Qwen backbones, the same backbones that dedicated rerankers such as Qwen3-Reranker [39] and Prism [38] are built on.

This makes a clean test possible, and the reports so far have not run it. They compare Jev with rerankers built on other, mostly smaller models, so they cannot tell how much of its effectiveness comes from decision fine-tuning and how much from the underlying language model. We therefore compare decision fine-tuning

with plain reranker fine-tuning of the same backbone. Calibration needs a test of its own. A ranking uses only the order of the scores, so a model can rank well with badly calibrated probabilities, and the other way round. Calibration matters when a score is read on an absolute scale: to decide how many passages to pass on to a language model, to drop passages that do not answer the query, or to estimate how many relevant passages there are. We therefore also test whether the advertised calibration holds for relevance and whether it helps in such uses:

RQ1 Do decision models rerank better than the untuned model or a reranker fine-tune of the same backbone?

RQ2 Are their relevance probabilities better calibrated than those of the untuned model and the reranker fine-tune, and is that calibration of any use?

Figure 1 shows how a decision model reranks and what we compare. For six Qwen backbones from 0.6B to 27B parameters, we evaluate the untuned model, asked zero-shot, its decision fine-tunes (Kev and Clef) and its reranker fine-tunes, including two we train ourselves on MS MARCO alone (at 9B as a control, at 27B where none has been released). All of them rerank the same BM25 candidates of TREC DL 2019–2023 and five BEIR collections, and all return a probability of relevance per passage. Decision and untuned models are asked whether the passage answers the query. For comparison, we repeat all their runs with the looser question of S1Rank [27], an earlier report on Jev as a reranker. We find:

(1) *Reranker fine-tuning beats decision fine-tuning.* From 0.6B to 9B, the reranker fine-tune ranks better on every backbone, by 5 to 29 points nDCG@10. The difference is significant on 48 of the 60 pairs of reranker fine-tune and collection, never significantly in favour of the decision fine-tune, and it holds on collections that no fine-tune was trained on. At 27B, the two tie. On no backbone does decision fine-tuning make the better reranker.
(2) *Up to 9B, decision fine-tuning adds nothing over the untuned model.* It stays within −2.5 and +0.7 points of simply asking the untuned model the same question, while reranker fine-tuning adds 6 to 30 points.
(3) *The calibration advantage depends on the question.* Asked whether a passage answers the query, decision models from 4B on are better calibrated than the published reranker fine-tunes. Asked S1Rank's question, they lose this advantage while their ranking hardly moves, and the untuned models gain as much from the wording. Calibration also does not make up for a worse ranking: up to 9B, a reranker fine-tune keeps the better passages at a fixed threshold.

If you need a reranker, fine-tune a reranker. Code, raw model outputs (a probability per query and passage) and all results are public.[1]

2 Related Work

LLM-based rerankers. Neural rerankers are trained to score query–passage pairs from relevance labels, first with encoder or sequence-to-sequence models [19, 20] and more recently with open large language models [17, 38, 39]. Qwen3-Reranker and Prism read relevance from the probability of "yes" against "no" after a prompt that asks whether the document meets the query, which makes their output directly comparable to that of a decision model. Other approaches rerank the candidates jointly, either by generating a ranked list [24, 31] or by letting passages attend to each other, as in the Set-Encoder [28]. Large language models can also rank without any reranking-specific fine-tuning, through relevance prompting [26, 40], but their effectiveness is sensitive to how the prompt is worded [30]. Abdallah et al. [1] compare many of these rerankers on the candidates we use.

Decision models in retrieval. Besides Jev [33], Kev [21, 22], Clef [4] and Laya [5] are open decision models. The evidence on decision models for retrieval is so far limited to reports that are not peer-reviewed: Jev reranking BM25 candidates [13, 15, 27], as a reranker in production search [11, 34], and the open Laya as a relevance gate before recall [5, 29]. These reports compare Jev with existing rerankers, which are built on other models. None compares decision fine-tuning with reranker fine-tuning of the same backbone, and none asks whether the calibration of the probabilities pays off.

Calibration of relevance scores. Neural networks tend to be overconfident, and temperature scaling after training is a simple remedy [12]. Neural rankers are no exception [23]. Common ranking losses are invariant to shifting the scores of a query, so rankers trained with them can order passages well without being calibrated [23, 36]; scores can be calibrated during or after training [12, 37]. Calibrated probabilities matter wherever a score is read on an absolute scale, for example to cut off a ranked list [2]. This is the use we test in RQ2.

3 Decision Models as Rerankers

A decision model takes an input, the *state*, together with one or more questions about it, and returns a probability for each possible answer. We use binary questions only, so every question yields one number, the probability of "yes". Reranking follows directly: for a query q with candidates $d_1, \ldots, d_k$ from a first-stage retriever, we ask about each passage d_i whether it answers q and order the passages by the returned probabilities p_i.

For the state, we consider two modes, the same as S1Rank [27]. In *isolated* mode, the state holds the query and a single passage, so that $p_i = P(yes \mid q, d_i)$, as in a cross-encoder. In *joint* mode, the state holds the query and all k candidates in first-stage order, and the k questions are asked about this one state, so that $p_i = P(yes \mid q, d_1, \ldots, d_k)$ can depend on the other candidates. Joint mode sees the same input as a listwise reranker such as RankGPT, but it returns a score per passage instead of an order, like the Set-Encoder [28]. In both modes the model scores each passage on its own scale, so both are pointwise rerankers.

Terms. A *backbone* is one pretrained Qwen model of a given size. For every backbone we compare three variants, all of which return a probability of relevance per query and passage:

- The *untuned model* is the released chat model of the backbone, not fine-tuned by anyone for ranking or for decisions. We ask it the decision models' question in its chat format and take p_i as the probability of "yes" as its first answer token.

[1] https://github.com/KanishkaGD/decision-rerank

- A *decision fine-tune* (Kev, Clef) is the backbone fine-tuned to act as a decision model.
- A *reranker fine-tune* (Qwen3-Reranker, Prism, ours) is the backbone fine-tuned on relevance labels. It keeps its own prompt with q and d_i and gives p_i as the probability of "yes" against "no".

We can therefore rank by p_i (RQ1) and read it on an absolute scale (RQ2) for every variant. *Decision models* are all models of this kind, including Jev (closed) and Laya, which we report for context only.

4 Experimental Setup

Backbones and variants. We study six Qwen backbones: Qwen3-0.6B and 8B, Qwen3.5-0.8B, 4B and 9B, and Qwen3.8-27B, each in the three variants of Sect. 3 (Figure 1). The untuned model is asked with thinking off. The decision fine-tunes are Kev [21, 22] on every backbone, Clef-Flash on the 9B and Clef on the 27B backbone [4], each run in isolated and in joint mode. Kev up to 9B is a LoRA fine-tune of the pretrained checkpoint; Kev-27B is a full fine-tune of the released chat model. The decision fine-tunes are trained to give Jev's kind of output; they are not reproductions of RLCD, which is unpublished [21]. The reranker fine-tunes are Qwen3-Reranker [39] (0.6B, 8B), built on the same pretrained checkpoints as Kev, and Prism [38] (0.8B, 4B, 9B).

Our reranker fine-tunes. No reranker fine-tune of Qwen3.8-27B has been released, so we train one ourselves (*Ours-27B*), from exactly the checkpoint that Kev-27B and Clef start from. As a control, we train a second one with the same recipe on Qwen3.5-9B (*Ours-9B*), where it can be compared with Prism-9B. The recipe is public and deliberately plain: LoRA ($r = 32$) in Qwen3-Reranker's prompt format, one epoch over 100k MS MARCO training queries [18], each with one relevant passage and 15 hard negatives (eight from a dense retriever, seven from BM25), and a ranking loss (InfoNCE over the group) plus a yes/no classification loss (binary cross-entropy of every passage's score). The second term gives the model a meaningful $P(\textit{yes})$. We removed training queries that match a test or development topic and chose the checkpoint on 200 MS MARCO dev queries by a rule fixed in advance. No decision data and no hyperparameter search are involved. Qwen3-Reranker, Prism and our fine-tunes saw MS MARCO in training; Kev's training data contain neither MS MARCO nor BEIR [21].

The question. Every decision and untuned model is asked one question per passage: "Does document `{doc}` answer the `query`?" It matches the relevance labels we evaluate against (on TREC DL, a passage counts as relevant if it answers the query, from grade 2). We fixed this question before running the experiments, but did not choose it on development data, and we report every result with it. The only other wording we ran is the question of S1Rank [27]: "Is document `{doc}` relevant to the `query`? Relevant means it answers the query or contains information that directly addresses it." Section 6 compares the two. Reranker fine-tunes ask no question of this kind; they keep their own prompt.

Collections and candidates. We use TREC DL 2019–2023 [6–10] on the MS MARCO v1 and v2 passages [18] and five BEIR collections [32]: TREC-COVID, NFCorpus, SciFact, Touché and DBPedia. None of the fine-tunes was trained on TREC-COVID, Touché or DBPedia, which form the *zero-shot subset* (ZS). Every system reranks the BM25 top-100 of Pyserini [16]; these are the candidates of S1Rank [27] and of earlier comparisons [1, 24, 31], whose published numbers we use where we did not rerun a system.

Other systems. For context, we report BM25, Jev 1.13.0 (closed, in both modes), Laya [5] (isolated only, as its context is too short for a joint state), monoT5-3B [20], bge-reranker-v2-m3 [3], the Set-Encoder [28], RankZephyr [24] (run by us on TREC DL) and RankGPT-4 [31] (published numbers only).

Ranking. We measure nDCG@10 and test differences with paired t-tests over queries, Holm-corrected across the ten collections ($p < 0.05$). Per backbone, we test three differences: the reranker fine-tune against the strongest decision fine-tune (the decision fine-tune and mode with the best average), that decision fine-tune against the untuned model, and joint against isolated mode.

Calibration and its uses. We compute the expected calibration error (ECE, 15 equal-width bins) [12] and the Brier score (the mean squared error of the probability against the binary label, which also rewards separating relevant from non-relevant passages) on the judged query–passage pairs of TREC DL 2019–2023, the only collections that also judge many non-relevant passages; a passage counts as relevant from grade 2. We test two uses of the probabilities. The *fixed cutoff* keeps every candidate with $p \geq 0.5$; we report the F_1 of this set (unjudged passages count as non-relevant) and its size. Its *regret* is what a better threshold would gain: the F_1 at the best threshold of each collection, chosen afterwards, minus the F_1 at 0.5. The *count test* adds up a query's probabilities over its judged passages. If the probabilities are calibrated, this sum estimates how many of the passages are relevant. We report the mean absolute error of this estimate, in passages. As a reference, always guessing the median number of relevant passages of the collection is off by 8.6 passages. Reranker fine-tunes are also calibrated after the fact, with temperature and Platt scaling of their logit fitted on 1,000 MS MARCO dev queries, never on a test collection.

Sanity checks. Our BM25, monoT5-3B and RankZephyr runs reproduce published numbers [1, 25] within 0.2 points nDCG@10. Jev in joint mode, asked S1Rank's question, reproduces S1Rank's numbers [27] within 0.5 points on four of five collections (1.5 below on TREC-COVID).

5 RQ1: Decision vs. Reranker Fine-Tuning

Reranker fine-tuning beats decision fine-tuning. On every backbone from 0.6B to 9B, the reranker fine-tune ranks better than the decision fine-tune in either mode (Table 1). Averaged over the ten collections, its margin over the better decision mode ranges from 5.2 points nDCG@10 at Qwen3.5-9B (Prism-9B 59.0 vs. Clef-Flash 53.8) to 28.8 points at Qwen3-0.6B (56.1 vs. 27.3). The difference is significant on 48 of the 60 pairs of reranker fine-tune and collection, and the decision fine-tune is never significantly better (*). The margin does not come from training on MS MARCO: on the zero-shot subset, which no fine-tune was trained on, it ranges from 5.4 to 23.9 points. Nor does it come from the undisclosed recipes of the published rerankers: Ours-9B, trained on MS MARCO alone with

Table 1: Untuned model, decision fine-tunes and reranker fine-tunes of the same backbone (nDCG@10, ×100, BM25 top-100). Untuned and decision models are asked whether the passage answers the query. Avg: mean over all ten collections; ZS: mean over the collections without a training split in any fine-tune's data (TREC-COVID, Touché, DBPedia). Per backbone and column: best in bold, second best underlined. Training: ○ untuned model, ▲/△ decision fine-tune (joint/isolated), ■ reranker fine-tune (rows shaded grey, red, blue); *Ours*: our reranker fine-tunes. Paired t-test, Holm-corrected over the ten collections, $p < 0.05$: * reranker fine-tune vs. the decision fine-tune with the best Avg, † that decision fine-tune vs. the untuned model, ‡ joint vs. isolated.

			TREC DL					BEIR					Mean	
	System	Mode	DL19	DL20	DL21	DL22	DL23	COVID	NFC	SciF	Touché	DBP	Avg	ZS
Qwen3-0.6B														
○	Qwen3-0.6B		43.3	32.7	37.7	22.1	17.9	50.9	15.1	14.2	16.8	15.3	26.6	27.7
▲	Kev-0.6B	joint	23.3‡	15.1‡	20.4‡	12.8‡	12.1‡	36.8‡	12.4‡	4.5‡	5.2‡	8.4‡	15.1	16.8
△	Kev-0.6B	isolated	35.3	30.5	40.1	22.8	19.0	55.9	16.5	15.5	14.1	22.9†	27.3	31.0
■	Qwen3-Reranker-0.6B		**69.5***	**64.0***	**65.2***	**46.4***	**38.8***	**86.2***	**36.4***	**76.6***	**35.7***	**42.7***	**56.1**	**54.9**
Qwen3-8B														
○	Qwen3-8B		67.3	63.7	62.0	38.3	39.0	77.6	32.7	73.3	25.5	31.6	51.1	44.9
▲	Kev-8B	joint	62.7	55.2	59.1	35.5	34.3	53.1‡	20.4‡	17.9‡	12.3‡	36.2‡	38.7	33.9
△	Kev-8B	isolated	69.0	63.1	61.9	38.6	36.4	80.6	33.8	69.8	25.2	39.0†	51.7	48.3
■	Qwen3-Reranker-8B		**73.2**	**66.8**	**67.0***	**51.2***	**45.2***	**84.0**	**37.3***	**79.9***	**43.1***	**46.2***	**59.4**	**57.8**
Qwen3.5-0.8B														
○	Qwen3.5-0.8B		46.0	41.2	52.1	28.1	27.9	59.2	18.3	27.0	16.4	22.3	33.9	32.6
▲	Kev-0.8B	joint	37.9	33.8	41.0	21.3	20.0	46.2‡	20.3	30.0	12.0	16.4‡	27.9	24.9
△	Kev-0.8B	isolated	41.8	38.4	47.9	24.7	23.9	68.1†	19.2	31.5	16.5	29.9†	34.2	38.2
■	Prism-0.8B		**72.3***	**67.7***	**66.3***	**47.0***	**43.7***	**84.2***	**36.1***	**76.3***	**35.6***	**41.7***	**57.1**	**53.8**
Qwen3.5-4B														
○	Qwen3.5-4B		67.9	64.0	65.9	42.3	40.4	78.0	33.3	74.1	25.0	33.0	52.4	45.3
▲	Kev-4B	joint	62.5	63.2	60.8	37.0	36.3	75.3	29.3‡	58.4‡	19.9	39.8	48.2	45.0
△	Kev-4B	isolated	62.7	62.9	60.9	37.9	34.5†	78.0	32.0	67.7†	23.4	39.0†	49.9	46.8
■	Prism-4B		**74.4***	**68.9***	**70.3***	**50.1***	**46.2***	**84.6***	**37.4***	**76.9***	**34.3***	**43.7***	**58.7**	**54.2**
Qwen3.5-9B														
○	Qwen3.5-9B		67.3	65.9	66.1	43.7	39.9	77.8	33.3	76.0	26.3	34.5	53.1	46.2
▲	Kev-9B	joint	62.2	62.3	60.5	37.7	36.5	76.8	31.4	56.0‡	23.7	36.1	48.3	45.5
△	Kev-9B	isolated	63.3	64.0	60.1	38.7	34.8	75.9	33.3	71.7	21.9	35.6	49.9	44.5
▲	Clef-Flash	joint	67.5	67.4	63.9	42.9	41.2	80.9	34.1‡	62.4‡	20.0	37.5‡	51.8	46.1
△	Clef-Flash	isolated	67.8	69.6†	64.5	43.8	43.1	80.5	36.1†	68.1†	23.4	40.8†	53.8	48.3
■	Prism-9B		**74.0***	**71.1**	**71.0***	51.6*	44.2	83.0	**37.2**	**79.8***	34.7*	43.2*	59.0	53.7
■	*Ours-9B*		73.8*	70.0	67.3	**52.1***	**47.0**	**84.2**	37.1	77.9*	**36.5***	**47.7***	**59.4**	**56.1**
Qwen3.8-27B														
○	Qwen3.8-27B		71.9	68.7	66.9	45.8	44.7	82.6	37.2	74.5	30.1	41.7	56.4	51.5
▲	Kev-27B	joint	72.8	**72.9**†	**70.7**†	50.9†	**49.2**†	**83.9**	**38.4**†	**81.1**†	**36.1**†	44.9†	**60.1**	**55.0**
△	Kev-27B	isolated	73.6	70.5	70.1	50.5	46.7	83.2	37.9	80.6	32.1	45.2	59.0	53.5
▲	Clef	joint	70.5	72.4	66.2	48.2	45.9	80.7	35.6	58.7	29.2	40.9	54.8	50.3
△	Clef	isolated	69.4	68.5	66.5	43.7	43.3	80.3	36.2	62.7	28.1	40.5	53.9	49.6
■	*Ours-27B*		**74.6**	69.6	69.0	**52.3**	48.3	83.0	38.0	79.2	34.6	**47.1***	59.6	54.9

standard losses, matches Prism-9B (59.4 vs. 59.0 on average, 56.1 vs. 53.7 on the zero-shot subset). At 27B, the two kinds of fine-tuning tie. Our reranker fine-tune of the checkpoint that Kev-27B starts from reaches 59.6 against Kev-27B's 60.1, and 54.9 against 55.0 on the zero-shot subset; they differ significantly on one collection only, DBPedia, where ours is better. On no backbone does decision fine-tuning make the better reranker.

Up to 9B, decision fine-tuning adds nothing over the untuned model. Up to 9B, the better mode of a decision fine-tune stays within −2.5 and +0.7 points of its untuned model, asked the same question, with significant gains on at most three of ten collections (†). Reranker fine-tuning of the same backbones adds 5.9 (Prism-9B) to 29.5 points (Qwen3-Reranker-0.6B). A user who takes the released Qwen3.5-4B or Qwen3.5-9B and asks it whether a passage answers the query gets the same ranking as from the decision fine-tunes built on it, at no training cost. At 27B, both kinds of fine-tuning add about the same to the untuned model (56.4): Kev-27B 3.7 points, significantly on eight collections, and our reranker fine-tune 3.2. Clef, the other 27B decision fine-tune, adds nothing (54.8 joint, 53.9 isolated).

Context. Table 2 places the triplets among other rerankers on the same candidates. Jev (59.1) ranks as well as the reranker fine-tunes of 8B and 9B (59.0 to 59.4) and the Set-Encoder (59.0). The open Kev-27B (60.1) matches the closed model it imitates, and so does our 27B reranker fine-tune (59.6). A fine-tune of a strong open backbone reaches Jev, with or without decision data. Laya (43.5) does not improve on its BM25 candidates (43.2).

Joint scoring helps only the largest models. Scoring all candidates in one state gains about one point for Kev-27B (60.1 vs. 59.0) and Clef (54.8 vs. 53.9), but costs Kev-0.6B 12.2 points and Kev-8B 13.0 (‡; significant on all ten and on five collections). A likely reason is the length of the state. A joint state of 100 passages has about 13k tokens on TREC DL and 40k or more on BEIR, whereas the model cards of the small Kevs report training states of at most 7,552 tokens (Kev-0.8B and Kev-4B), and that of Kev-27B states of up to 32,768 [21].

Table 2: Context: other rerankers (nDCG@10, ×100) on the BM25 top-100 of TREC DL 2019–2023 and five BEIR collections; System One models in joint mode (Laya isolated), asked whether the passage answers the query (Sect. 4). †: published number on the same candidates [1, 35], not recomputed. Avg: mean over all ten collections (only where all are available). $/1k: mean API cost in US dollars per 1,000 queries over the collections run. Best reranker per column in bold.

		TREC DL					BEIR						
System	Mode	DL19	DL20	DL21	DL22	DL23	COVID	NFC	SciF	Touché	DBP	Avg	$/1k
BM25		50.6	48.0	44.6	26.9	26.2	59.5	32.2	67.9	44.2	31.8	43.2	–
Listwise LLM rerankers													
RankGPT-4		**75.6**†	70.6†	–	–	–	**85.5**†	38.5†	75.0†	38.6†	**47.1**†	–	–
RankZephyr-7B		74.3	70.0	67.9	50.5	44.1	80.7†	36.6†	75.1†	31.1†	43.2†	57.3	–
Cross-encoders and LLM rerankers													
bge-reranker-v2-m3		72.2†	67.0†	66.1	51.5	45.1	74.8†	33.8†	73.5†	39.9†	41.9†	56.6	–
monoT5-3B		71.8	68.9	66.7	49.5	45.0	80.7†	**39.0**†	76.6†	32.4†	44.5†	57.5	–
Qwen3-Reranker-8B		73.2	66.8	67.0	51.2	45.2	84.0	37.3	79.9	**43.1**	46.2	59.4	–
Set-Encoder-large		75.0†	72.2†	69.9	**53.4**	47.7	81.2	36.8	73.8	33.2	47.0	59.0	–
System One models													
Jev	joint	73.0	70.9	**70.7**	50.3	46.1	83.3	37.8	80.3	33.5	45.5	59.1	1.11
Clef-Flash	joint	67.5	67.4	63.9	42.9	41.2	80.9	34.1	62.4	20.0	37.5	51.8	–
Clef	joint	70.5	72.4	66.2	48.2	45.9	80.7	35.6	58.7	29.2	40.9	54.8	–
Kev-4B	joint	62.5	63.2	60.8	37.0	36.3	75.3	29.3	58.4	19.9	39.8	48.2	–
Kev-27B	joint	72.8	**72.9**	**70.7**	50.9	**49.2**	83.9	38.4	**81.1**	36.1	44.9	**60.1**	–
Laya	isolated	59.5	54.6	52.8	35.3	32.7	73.8	28.7	44.8	21.8	31.3	43.5	–

6 RQ2: Are Decision Models Better Calibrated?

Calibration is the main promise of decision models. Table 3 reports it on the judged pairs of TREC DL 2019–2023, together with the two uses we test. Table 4 repeats every run of a decision or untuned model with S1Rank's question.

Asked our question, decision models are well calibrated. The untuned models of 0.6B and 0.8B parameters say "yes" to almost everything (ECE 0.750 and 0.617); from 4B on, they are reasonably calibrated already (0.117 to 0.231). Decision fine-tuning improves on the untuned model on every backbone, for example from 0.231 to 0.113 on Qwen3-8B. From 4B on, the decision fine-tunes are also better calibrated than the published reranker fine-tunes (Kev-8B 0.113 vs. Qwen3-Reranker-8B 0.257); on the two smallest backbones they are not. The best calibrated systems in the study are Jev (0.075) and Kev-27B (0.076). Decision fine-tuning is not the only way there: our 9B reranker fine-tune, trained with a plain yes/no loss, is as well calibrated as Clef-Flash (0.111 vs. 0.112). Our 27B reranker fine-tune is less well calibrated than Kev-27B (0.110).

The advantage depends on the question. S1Rank's question asks whether the passage is relevant and defines relevance more loosely: the passage "answers the query or contains information that directly addresses it". Both wordings are reasonable, and S1Rank's is the one under which Jev was first shown to be a good reranker. The ranking hardly moves between the two: the mean nDCG@10 of the decision fine-tunes changes by −2.5 to +2.6 points, and with S1Rank's question, too, the best decision fine-tune of every backbone up to 9B is 4 points or more behind the reranker fine-tune. The answer to RQ1 does not depend on the wording. Calibration moves a lot. With S1Rank's question, almost every model is worse calibrated, often by a factor of two or three (Jev joint: ECE 0.224 instead of 0.075). The decision fine-tunes are then at best as well calibrated as the best reranker fine-tune of their backbone (Kev-8B 0.243 vs. Qwen3-Reranker-8B 0.257, Kev-27B 0.156 vs. Ours-27B 0.110). One sentence of the prompt decides whether decision models are better calibrated than reranker fine-tunes.

The untuned models gain as much from our wording as the decision fine-tunes (Qwen3-8B: 0.383 with S1Rank's question, 0.231 with ours). A question that matches the definition of relevance in the labels makes the probabilities better calibrated, with or without decision fine-tuning. The user of a decision model has to put that definition into words, and a small change in the words changes the calibration severalfold. A reranker fine-tune has no such knob: its notion of relevance is fixed by its training labels.

Calibration does not make up for a worse ranking. If every passage with $p \geq 0.5$ is kept, a reranker fine-tune keeps the better set on every backbone up to 9B (e.g. F_1@.5 of 51.0 for Prism-4B vs. 42.6 for Kev-4B), although it is the worse calibrated model on most of them. The threshold is not the problem: for the decision models, 0.5 is close to the best threshold of each collection (regret of 0.9 to 3.1 points). A well-placed threshold on a worse ranking still yields the worse set. Where the rankings are equally good, calibration does pay off: at 27B, the better calibrated Kev-27B keeps the better set (57.2 vs. 48.3), while our fine-tune is underconfident and keeps only 7.9 passages per query.

Counting relevant passages. A ranking cannot tell how many passages are relevant; the sum of calibrated probabilities can. Asked our question, most decision models from 4B on count better than the median guess (8.6 passages off) and better than the published reranker fine-tunes of their backbone (Kev-8B 7.6 vs. Qwen3-Reranker-8B 12.9). Decision fine-tuning is not what makes this work: the untuned Qwen3.8-27B counts most accurately of all (5.8), and Ours-9B counts better than Clef-Flash (7.3 vs. 8.1). With S1Rank's question, no decision model beats the median guess by more than a fraction of a passage (best: Clef joint, 8.4).

Calibration after the fact. Rerankers can also be calibrated after training. Temperature scaling fitted on MS MARCO dev lowers the ECE of the worst calibrated rerankers (Qwen3-Reranker-0.6B from

Table 3: Calibration and its uses on the judged pairs of TREC DL 2019–2023 (means). Per backbone: untuned model, decision fine-tunes in their better-calibrated mode (j/i: joint/isolated), reranker fine-tunes. ECE (lower is better): N native; T, P after temperature/Platt scaling fitted on MS MARCO dev (rerankers only). Brier: mean squared error of the probability. F_1@.5 (×100): set kept at $p \geq 0.5$, native (N) and Platt-scaled (P); Regret: F_1 at each collection's best cutoff minus F_1@.5; $|S|$: passages kept at 0.5 (of 100). Count: mean absolute error of the summed probabilities against the number of relevant passages (always guessing the median: 8.6). Per backbone and column: best in bold, second best underlined.

		ECE				F_1@.5				
	System	N	T	P	Brier	N	P	Regret	$\|S\|$	Count
Qwen3-0.6B										
○	Qwen3-0.6B	0.750	–	–	0.750	19.9	–	**0.5**	100.0	36.5
△	Kev-0.6B (i)	**0.503**	–	–	**0.463**	21.4	–	2.2	79.8	**24.7**
■	Qwen3-Reranker-0.6B	0.511	**0.326**	**0.226**	0.481	**31.8**	**6.3**	14.9	58.4	24.8
Qwen3-8B										
○	Qwen3-8B	0.231	–	–	0.233	42.5	–	**2.1**	22.9	8.7
▲	Kev-8B (j)	**0.113**	–	–	**0.177**	34.0	–	3.1	13.7	**7.6**
■	Qwen3-Reranker-8B	0.257	**0.237**	**0.227**	0.238	**46.3**	**7.4**	9.0	32.2	12.9
Qwen3.5-0.8B										
○	Qwen3.5-0.8B	0.617	–	–	0.566	21.6	–	11.2	88.8	29.9
△	Kev-0.8B (i)	0.336	–	–	0.297	25.1	–	2.9	51.9	16.9
■	Prism-0.8B	**0.243**	**0.229**	**0.237**	**0.193**	**46.9**	**5.7**	**2.8**	26.5	**12.7**
Other systems										
▲	Jev (j)	**0.075**	–	–	**0.127**	**54.7**	–	1.7	16.0	**6.3**
△	Laya	0.127	–	–	0.183	37.5	–	**1.0**	23.3	9.2
■	monoT5-3B	0.312	0.257	**0.232**	0.295	42.0	**5.8**	9.9	35.6	15.2
■	bge-reranker-v2-m3	0.231	**0.226**	0.236	0.228	48.2	5.7	3.8	29.3	12.1
■	Set-Encoder-large	0.495	0.250	0.236	0.426	31.6	5.7	21.1	52.2	24.3

		ECE				F_1@.5				
	System	N	T	P	Brier	N	P	Regret	$\|S\|$	Count
Qwen3.5-4B										
○	Qwen3.5-4B	0.181	–	–	0.195	44.3	–	3.1	27.1	9.8
△	Kev-4B (i)	**0.132**	–	–	0.175	42.6	–	**1.4**	21.9	**8.4**
■	Prism-4B	0.218	**0.202**	**0.238**	**0.172**	**51.0**	**5.7**	2.8	23.1	11.5
Qwen3.5-9B										
○	Qwen3.5-9B	0.165	–	–	0.181	46.5	–	2.9	24.4	8.7
▲	Kev-9B (j)	0.160	–	–	0.177	40.2	–	2.3	23.8	9.5
▲	Clef-Flash (j)	0.112	–	–	0.158	47.1	–	**1.7**	18.9	8.1
■	Prism-9B	0.242	0.233	0.238	0.183	**50.7**	5.7	4.0	26.0	12.3
■	*Ours-9B*	**0.111**	**0.108**	**0.231**	**0.157**	46.6	**6.9**	6.4	8.7	**7.3**
Qwen3.8-27B										
○	Qwen3.8-27B	0.117	–	–	0.144	54.1	–	1.9	16.0	**5.8**
▲	Kev-27B (j)	**0.076**	–	–	**0.130**	**57.2**	–	**0.9**	16.1	6.0
▲	Clef (j)	0.091	–	–	0.139	53.4	–	2.2	17.7	6.7
■	*Ours-27B*	0.110	**0.114**	**0.230**	0.153	48.3	**8.1**	5.9	7.9	7.3

0.511 to 0.326) and changes the others little. Platt scaling backfires at the fixed cutoff (F_1@.5 of 8.1 or less): MS MARCO dev has about one relevant passage per query, so the fitted probabilities are far too low for TREC DL, where queries have many more. A calibration holds only for the notion of relevance and the base rate it was fitted on (Sect. 7).

7 Discussion and Limitations

Why decision fine-tuning does not make a reranker. A reranker fine-tune is trained on the very task it is tested on: telling relevant from non-relevant passages of a query, with a ranking loss over hard negatives. The decision fine-tunes we tested learned to answer many kinds of questions about many kinds of states, and Kev's training data contain no MS MARCO and no BEIR [21]. That general skill transfers to relevance about as well as the untuned model's own instruction following, which is what we observe up to 9B. The zero-shot subset shows that the gap is not about having seen the test domain: it persists on collections that neither kind of fine-tune was trained on.

Why 27B is different. Kev-27B is the only decision fine-tune that adds to its untuned model. It differs from the smaller Kevs in more than size: it is a full fine-tune of the released chat model, whereas the smaller Kevs are LoRA fine-tunes of pretrained checkpoints, and it was trained on longer states (Sect. 5). We cannot tell which of these matters. Size alone is not enough: Clef, a decision fine-tune of the same 27B model, adds nothing. Reranker fine-tuning also adds less at 27B than on the smaller backbones, because the untuned 27B model already ranks well.

What calibration means for relevance. A probability of relevance is calibrated only with respect to a definition of relevance and its base rate. Our results show three ways to move it: the wording of the question (Sect. 6), the labels of the training data (our reranker fine-tunes) and the labels a scaling is fitted on (Platt scaling on MS MARCO dev, Sect. 6). A model that is "calibrated out of the box" is calibrated against a notion of relevance that someone chose, and the user of a decision model has to put that notion into words. For ranking none of this matters; for a threshold or a count it matters a lot.

Limitations. We compare released models, and the fine-tunes of a backbone differ in their training data as well as in their objective. Our results therefore describe decision fine-tuning as it is available today, not RLCD as such, which is unpublished and may be done better. The controlled comparison rests on our own reranker fine-tunes at 9B and 27B, which were trained once, with one recipe and without any hyperparameter search. We measured calibration only on TREC DL, where non-relevant passages are judged, and at a single relevance grade and a single threshold. Our question was fixed before the runs but not chosen on development data, and we tried only one other wording. All decision models are at most a month old, and newer versions may change the picture; we release all raw outputs so that the comparison can be repeated.

8 Conclusion

System One models are not better rerankers. On the same backbone, decision fine-tuning never makes the better reranker: plain reranker fine-tuning wins by 5 to 29 points nDCG@10 from 0.6B to 9B, and the two tie at 27B. Up to 9B, decision fine-tuning adds nothing over simply asking the untuned model. The calibration advantage of decision models depends on the question. A question that matches the relevance labels makes decision and untuned models alike much better calibrated; with a looser one, decision models lose their advantage over reranker fine-tunes. Up to 9B, calibration also does not make up for a worse ranking when passages are kept at a fixed threshold. If you need a reranker, fine-tune a reranker.

Table 4: The question ablation: S1Rank's question (*Rel.*: "Is document {doc} relevant to the query? Relevant means it answers the query or contains information that directly addresses it.") against the question used in the rest of the paper (*Ans.*: "Does document {doc} answer the query?"), for every system that is asked a question. nDCG@10 (×100): mean over all ten collections (Avg) and over TREC DL 2019–2023 (DL); ECE and Count as in Table 3 (DL 2019–2023). Δ: Ans. minus Rel. Training: ○ untuned model, ▲/△ decision fine-tune (joint/isolated), ■ reranker fine-tune.

			nDCG@10 Avg			nDCG@10 DL			ECE			Count		
	System	Mode	Rel.	Ans.	Δ	Rel.	Ans.	Δ	Rel.	Ans.	Δ	Rel.	Ans.	Δ
Qwen3-0.6B														
○	Qwen3-0.6B		23.8	26.6	+2.8	25.4	30.7	+5.3	0.750	0.750	0.000	36.5	36.5	0.0
▲	Kev-0.6B	joint	13.2	15.1	+1.9	14.9	16.7	+1.8	0.591	0.503	−0.088	28.8	24.6	−4.2
△	Kev-0.6B	isolated	27.0	27.3	+0.3	28.8	29.5	+0.7	0.579	0.503	−0.076	28.2	24.7	−3.5
Qwen3-8B														
○	Qwen3-8B		53.3	51.1	−2.2	56.9	54.0	−2.9	0.383	0.231	−0.152	17.5	8.7	−8.8
▲	Kev-8B	joint	38.8	38.7	−0.1	50.9	49.3	−1.6	0.243	0.113	−0.130	12.8	7.6	−5.2
△	Kev-8B	isolated	52.7	51.7	−1.0	54.2	53.8	−0.4	0.283	0.154	−0.129	14.1	8.7	−5.4
Qwen3.5-0.8B														
○	Qwen3.5-0.8B		37.6	33.9	−3.7	40.3	39.1	−1.2	0.716	0.617	−0.099	34.8	29.9	−4.9
▲	Kev-0.8B	joint	25.3	27.9	+2.6	28.9	30.8	+1.9	0.395	0.398	+0.003	19.5	19.6	+0.1
△	Kev-0.8B	isolated	36.0	34.2	−1.8	36.1	35.3	−0.8	0.357	0.336	−0.021	18.0	16.9	−1.1
Qwen3.5-4B														
○	Qwen3.5-4B		52.7	52.4	−0.3	55.8	56.1	+0.3	0.296	0.181	−0.115	14.7	9.8	−4.9
▲	Kev-4B	joint	49.8	48.2	−1.6	53.4	51.9	−1.5	0.251	0.201	−0.050	13.2	11.6	−1.6
△	Kev-4B	isolated	51.3	49.9	−1.4	52.7	51.8	−0.9	0.216	0.132	−0.084	11.5	8.4	−3.1
Qwen3.5-9B														
○	Qwen3.5-9B		53.7	53.1	−0.6	56.7	56.6	−0.1	0.283	0.165	−0.118	14.1	8.7	−5.4
▲	Kev-9B	joint	50.2	48.3	−1.9	54.0	51.9	−2.1	0.237	0.160	−0.077	12.2	9.5	−2.7
△	Kev-9B	isolated	50.1	49.9	−0.2	52.8	52.2	−0.6	0.348	0.221	−0.127	16.9	11.8	−5.1
▲	Clef-Flash	joint	52.8	51.8	−1.0	57.0	56.6	−0.4	0.140	0.112	−0.028	8.9	8.1	−0.8
△	Clef-Flash	isolated	54.7	53.8	−0.9	58.9	57.8	−1.1	0.173	0.144	−0.029	9.7	8.2	−1.5
Qwen3.8-27B														
○	Qwen3.8-27B		57.4	56.4	−1.0	59.9	59.6	−0.3	0.166	0.117	−0.049	8.9	5.8	−3.1
▲	Kev-27B	joint	61.1	60.1	−1.0	63.5	63.3	−0.2	0.156	0.076	−0.080	8.5	6.0	−2.5
△	Kev-27B	isolated	59.8	59.0	−0.8	62.5	62.3	−0.2	0.262	0.180	−0.082	12.9	9.4	−3.5
▲	Clef	joint	56.1	54.8	−1.3	61.1	60.6	−0.5	0.147	0.091	−0.056	8.4	6.7	−1.7
△	Clef	isolated	56.4	53.9	−2.5	58.5	58.3	−0.2	0.240	0.153	−0.087	12.1	8.3	−3.8
Other systems														
▲	Jev	joint	60.5	59.1	−1.4	62.7	62.2	−0.5	0.224	0.075	−0.149	11.3	6.3	−5.0
△	Jev	isolated	59.5	57.1	−2.4	61.4	60.0	−1.4	0.260	0.125	−0.135	12.9	7.8	−5.1
△	Laya	isolated	44.0	43.5	−0.5	47.3	47.0	−0.3	0.131	0.127	−0.004	9.3	9.2	−0.1

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